

Sprint Planning

Sprint 1 – Project Setup

Sprint Goal

Establish the project foundation by defining the game concept, preparing the product backlog, and configuring the development environment and collaboration tools.

Sprint Length

2 weeks (February 21 – March 6)

Sprint Backlog

1. PB-01 Project Environment Setup
 - Create Unity project
 - Create GitHub repository
 - Verify Unity project compatibility across Mac and Windows systems
2. PB-02 Version Control Configuration
 - Research GitHub large file handling for Unity assets
 - Configure Git LFS for Unity asset management
3. PB-03 Game Concept Definition
 - Discuss and finalize initial game concept
 - Define core gameplay mechanics
4. PB-04 Backlog and Project Board Setup
 - Create initial product backlog
 - Write user stories
 - Create GitHub Kanban project board
 - Organize backlog items on the Kanban board
5. PB-05 Asset Research
 - Research and evaluate character sprite assets
 - Research and evaluate world/environment assets

Daily Scrum Notes

Meeting	Notes
1	Discussed game concept and core gameplay mechanics. Identified necessary development tools and workflow. Assigned responsibilities for backlog creation and asset research.
2	Configured Unity project and GitHub repository. Tested Unity project compatibility across Mac and Windows systems. Researched Git large file handling and configured Git LFS. Created backlog and GitHub Kanban board to track tasks.

Sprint Review

During Sprint 1 the team successfully established the technical and planning foundation for the project. The Unity project and GitHub repository were created and configured to support collaboration across Mac and Windows systems. Git large file handling was researched and configured to support Unity asset management. The team also defined the initial game concept, created the product backlog and user stories, and set up a GitHub Kanban board to manage development tasks. These activities prepared the project for gameplay feature development in future sprints.

Retrospectives

1. What went well

- The development environment and version control were successfully configured.
- The team established a clear game concept and gameplay direction.
- The product backlog and Kanban board helped organize upcoming work.

2. What didn't go well

- Environment setup and Git configuration took longer than expected.
- Some time was spent researching tools rather than implementing gameplay features.

3. What can be improved

- Plan technical setup tasks earlier in the sprint.
- Begin implementing small gameplay features sooner.

4. Action Items / Next Steps

- Begin implementing player movement mechanics.
- Import selected assets into the Unity project.
- Start development of core gameplay systems.

Sprint Progress

Sprint 1 focused on planning, environment setup, and project organization rather than gameplay implementation. The development environment, version control system, and collaboration tools were successfully configured. The product backlog and user stories were created, and asset research was completed. These preparations provide a strong foundation for implementing gameplay features in the next sprint.

Sprint 2 – Core Movement and Environment

Sprint Goal

Create the first playable prototype by implementing player movement mechanics, forward progression, and obstacle interaction.

Sprint Length

2 weeks (March 7 – March 20)

Sprint Backlog

1. PB-06 Vertical Player Movement Control
 - Implement vertical player movement control script
 - Test responsiveness of vertical player movement controls
2. PB-07 Flap / Lift Mechanic
 - Implement flap / lift input mechanic
 - Tune upward force values for flap mechanic
3. PB-08 Gravity System
 - Apply gravity physics to player character
 - Test gravity behaviour during gameplay
4. PB-10 Horizontal Auto-Movement
 - Implement automatic forward player movement
 - Test horizontal movement speed consistency
5. PB-11 Obstacle Spawning System
 - Create obstacle prefabs for environment hazards
 - Implement obstacle spawn timing system
 - Test obstacle spawning and spacing during gameplay
6. PB-12 Collision Damage System
 - Implement collision detection between player and obstacles
 - Apply damage to player when collisions occur

Daily Scrum Notes

Meeting	Notes
1	Discussed and agreed on player flap movement. Implemented vertical and horizontal movement. Added flap lift and needed animation to simulate a flapping bird.
2	Gravity system implemented. Play scene auto moves horizontally infinitely
3	The obstacles are now able to appear infinitely at random positions (from our generator). Collision with the pipes now results in immediate death.

Sprint Review

During Sprint 2 the team successfully implemented basic player movement with realistic physics. In addition to that, a good gravity system and flap mechanism was implemented so that players can move through the obstacles. As the player moves, the scene auto moves horizontally in parallel with the autogenerated pipes(obstacles). The obstacles are made so that when the player collides with them, it immediately dies regardless of the amount of heart it had.

Retrospectives

1. What went well

- Core gameplay mechanics were successfully implemented
- A playable prototype was achieved by the end of the sprint

2. What didn't go well

- Limited communication and collaboration during the sprint due to midterms
- Obstacle spawning system is in place, but not yet optimal

3. What can be improved

- Improve communication consistency between team members
- Break tasks into smaller steps to better track progress

4. Action Items / Next Steps

- Refine obstacle spawning system (timing and positioning)
- Begin implementation of enemy and combat systems
- Improve coordination and task distribution for next sprint

Sprint Progress

Sprint 2 focused on implementing the core gameplay systems required for a playable prototype. Most planned features were completed, including player movement, physics, and collision systems. The obstacle spawning system was partially implemented but requires further refinement. Progress during the sprint was impacted by midterm exams and reduced team availability; however, with midterms completed, improved productivity and communication are expected in the next sprint.

Sprint 3 – Combat and Enemy Systems

Sprint Goal

Introduce enemy interactions and combat mechanics to expand gameplay challenge.

Sprint Length

1 week (March 21 – March 27)

Sprint Backlog

1. PB-13 Basic Attack Mechanic
 - Implement player attack input
 - Create attack animation or attack effect
2. PB-14 Enemy Health System
 - Implement enemy health tracking system
 - Apply damage to enemies when attacked
3. PB-15 Stationary Enemy Type
 - Create stationary enemy prefab
 - Implement stationary enemy collision behaviour
4. PB-16 Patrol Enemy Type
 - Implement patrol enemy movement pattern
 - Test patrol enemy interaction with player
5. PB-17 Aggressive Dive Enemy
 - Implement aggressive dive enemy behaviour
 - Test enemy dive attack behaviour
6. PB-21 Hit Feedback
 - Add visual or particle feedback when enemies are hit

Daily Scrum Notes

Meeting	Notes

Sprint Review

During Sprint x ..

Retrospectives

1. What went well
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2. What didn't go well

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3. What can be improved

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4. Action Items / Next Steps

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Sprint Progress

Sprint x ...

Sprint 4 – Gameplay Flow and UI

Sprint Goal

Complete gameplay systems and polish the player experience with scoring, user interface, and difficulty balancing.

Sprint Length

1 week (March 28 – April 3)

Sprint Backlog

1. PB-18 Score System
 - Implement score tracking system
 - Increase score when enemies are defeated
 - Increase score when obstacles are passed
2. PB-19 Health Display HUD
 - Implement health display HUD
 - Update health display when damage occurs
3. PB-20 Game Over Screen
 - Implement game over condition when health reaches zero
 - Create game over screen UI
4. PB-22 Difficulty Scaling
 - Implement difficulty scaling system
 - Adjust spawn rates or enemy difficulty over time
5. PB-23 Entry Screen
 - Create entry/start screen for the game
 - Implement start button to begin gameplay
6. PB-09 Glide Control Option
 - Implement hold-to-glide mechanic
 - Tune glide lift values for gameplay balance
7. PB-24 Power Ups
 - Implement collectible power-up objects
 - Apply temporary power-up effects to player

Daily Scrum Notes

Meeting	Notes

Sprint Review

During Sprint x ..

Retrospectives

1. What went well

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2. What didn't go well

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3. What can be improved

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4. Action Items / Next Steps

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Sprint Progress

Sprint x ...